Summary of Appliance Models Used

Mini Project 3: Harmonic Signature Disaggregation for Load Monitoring on an Unstable Feeder

This document summarizes the four appliance current models used in the clean aggregate signal simulation. Each appliance was assigned a different physical load type, RMS current level, phase angle, harmonic signature, on/off timing, and startup behavior so that the appliances can be distinguished using FFT-based harmonic analysis.

Appliance	Load type	Nominal RMS current	Main identifying signature
Electric kettle / heater	Resistive heating load	6.0 A	Strong 50 Hz fundamental, very low harmonics
Refrigerator compressor motor	Inductive motor load	2.0 A	Lagging phase, moderate harmonics, startup inrush
LED lamp bank / rectifier driver	Nonlinear rectifier lighting load	0.7 A	Strong odd harmonics, especially 3rd, 5th, and 7th
Laptop charger / SMPS	Switch-mode power supply load	1.1 A	Strong 5th, 7th, 11th, 13th, and 15th harmonics

1. Electric Kettle / Resistive Heater

This appliance represents a mostly resistive load such as a kettle, iron, or heating element. Its current is almost in phase with the supply voltage and is close to a pure sine wave.

Nominal RMS current: 6.0 A; fundamental phase: 0 degrees.

Harmonic	Magnitude ratio	Phase
1st	1.00	0 deg
3rd	0.020	20 deg
5th	0.010	-35 deg

ON intervals: 1.0 s to 5.5 s, and 8.5 s to 11.5 s.

Startup behavior: short soft ramp.

Main identifying feature: large fundamental current with very small harmonic distortion.

2. Refrigerator Compressor Motor

This appliance represents an inductive motor load. Its current lags the supply voltage, making it different from the resistive heater.

Nominal RMS current: 2.0 A; fundamental phase: -35 degrees.

Harmonic	Magnitude ratio	Phase
1st	1.00	-35 deg
3rd	0.080	-80 deg
5th	0.050	110 deg
7th	0.025	40 deg

ON interval: 2.5 s to 10.0 s.

Startup behavior: motor inrush, with an initial current surge about 3.2 times the normal value before settling.

Main identifying feature: lagging phase plus startup inrush.

3. LED Lamp Bank / Rectifier Driver

This appliance represents a nonlinear rectifier-based lighting load. Although its RMS current is low, it has strong harmonic content.

Nominal RMS current: 0.7 A; fundamental phase: 5 degrees.

Harmonic	Magnitude ratio	Phase
1st	1.00	5 deg
3rd	0.55	-10 deg
5th	0.35	60 deg
7th	0.22	-50 deg
9th	0.14	20 deg
11th	0.10	-90 deg

ON intervals: 0.5 s to 3.5 s, 4.8 s to 9.2 s, and 10.0 s to 12.0 s.

Startup behavior: very quick soft ramp.

Main identifying feature: very strong 3rd harmonic and visible odd harmonics.

4. Laptop Charger / Switch-Mode Power Supply

This appliance represents a switch-mode power supply such as a laptop charger. Its signature differs from the LED load because the 5th and 7th harmonics are strong and higher harmonics are included.

Nominal RMS current: 1.1 A; fundamental phase: 18 degrees.

Harmonic	Magnitude ratio	Phase
1st	1.00	18 deg
3rd	0.25	-40 deg
5th	0.50	70 deg
7th	0.40	-110 deg
11th	0.26	130 deg
13th	0.18	-70 deg
15th	0.10	30 deg

ON interval: 6.0 s to 12.0 s.

Startup behavior: charger inrush, with a short boost at turn-on before settling.

Main identifying feature: strong 5th, 7th, and higher odd harmonics.

Why the four appliances are distinguishable

The kettle is mainly identified by high fundamental current and very low harmonics. The motor is identified by phase lag and startup inrush. The LED driver is identified by a strong 3rd harmonic and other odd harmonics. The laptop charger is identified by stronger 5th, 7th, and higher odd harmonics. These differences make the four appliance signals separable using FFT harmonic magnitude ratios, phase information, and startup behavior.

Aggregate construction note

All appliance current vectors are generated using the same sampling frequency, 50 Hz fundamental frequency, and common time vector. The clean aggregate current is formed by adding the four current vectors sample by sample. Ground-truth ON/OFF states are stored both per sample and per analysis window for later classification and confusion-matrix evaluation.